Orthogonality of Irreducible Characters

An End-to-End Meta-Math Walkthrough

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Abstract

Average the group action on the space of all linear maps. Its image consists of intertwiners, and its trace can also be computed from the two characters. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RT-03 / Representation Theory and Lie Theory / Report v1

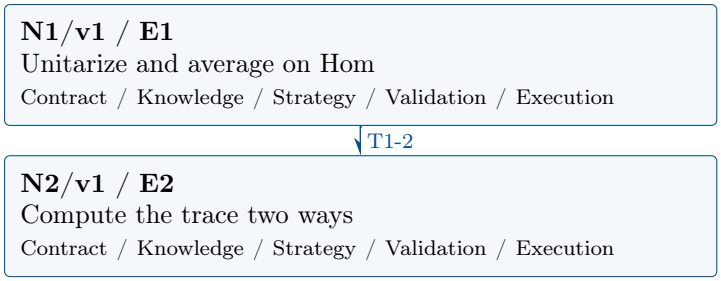
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let G be a finite group and V, W finite-dimensional irreducible complex representations. For $\chi_V(g) = \operatorname{tr} \rho_V(g)$, prove $|G|^{-1} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = 1$ if $V \cong W$ and 0 otherwise.*

A. Foundations and permitted imports

Import RT-02: the complex vector space $\mathrm{Hom}_G(V, W)$ has dimension one or zero in the stated cases. Use finite-dimensional trace, rank of an idempotent, and positive-definite Hermitian forms. An idempotent decomposes its space into image and kernel and has trace equal to image dimension; this is justified in the proof.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $P(A) = |G|^{-1} \sum_g g \cdot A$ is a projection onto $\mathrm{Hom}_G(V, W)$, and $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Receiving activity. N2.1

Compatibility. E1/v1 and the Hom dimension from RT-02. Check: Character conjugation belongs on V, the representation appearing with an inverse in the Hom action.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Unitarize and average on Hom

Object and role. Average positive forms to compare inverse traces with conjugates, and average the Hom action to project onto invariants. **Local obligation:** $P(A) = |G|^{-1} \sum_g g \cdot A$ is a projection onto $\mathrm{Hom}_G(V, W)$, and $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Inputs. $\mathrm{Hom}_{\mathbb{C}}(V, W)$ with $g \cdot A = \rho_W(g) A \rho_V(g)^{-1}$.

C — Contract

Publish E1/v1 only after establishing this obligation: $P(A) = |G|^{-1} \sum_g g \cdot A$ is a projection onto $\mathrm{Hom}_G(V, W)$, and $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

K — Knowledge

$\mathrm{Hom}_{\mathbb{C}}(V, W)$ with $g \cdot A = \rho_W(g) A \rho_V(g)^{-1}$. Retain the stated assumptions and import versions.

S — Strategy

Average positive forms to compare inverse traces with conjugates, and average the Hom action to project onto invariants.

V — Validation

Every summand of the averaged form is positive on a nonzero vector; finiteness makes the sum well defined.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Start with a positive Hermitian form on each representation and replace it by the sum $\sum_g \langle \rho(g)v, \rho(g)w \rangle$. This remains positive definite and is invariant by reindexing. In orthonormal bases every $\rho(g)$ is unitary, so the trace of its inverse is the complex conjugate of its trace.

N1.2 The conjugation formula defines a representation on Hom. Reindexing the average shows $P(A)$ is fixed by every group element. A fixed A is unchanged by averaging; fixedness is exactly $A\rho_V(g) = \rho_W(g)A$. Therefore $P^2 = P$ and its image is the intertwiner space.

Export E1/v1. $P(A) = |G|^{-1} \sum_g g \cdot A$ is a projection onto $\mathrm{Hom}_G(V, W)$, and $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Acceptance gate. Every summand of the averaged form is positive on a nonzero vector; finiteness makes the sum well defined.

Targeted repair. If an arbitrary initial inner product is assumed invariant, perform the finite-group averaging before identifying inverse and conjugate traces. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Compute the trace two ways

Object and role. Use a matrix-unit basis to factor the trace of left-right multiplication. **Local obligation:** The normalized character inner product equals $\dim \mathrm{Hom}_G(V, W)$ and hence the claimed Kronecker value.

Inputs. E1/v1 and the Hom dimension from RT-02. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: The normalized character inner product equals $\dim \mathrm{Hom}_G(V, W)$ and hence the claimed Kronecker value.

K — Knowledge

E1/v1 and the Hom dimension from RT-02. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Use a matrix-unit basis to factor the trace of left-right multiplication.

V — Validation

Character conjugation belongs on V, the representation appearing with an inverse in the Hom action.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 For matrices B on W and C on V, the diagonal coefficient of $A \mapsto BAC$ on the matrix unit E_{ij} is $B_{ii}C_{jj}$. Summing over i, j yields $\mathrm{tr}(A \mapsto BAC) = \mathrm{tr}\, B\, \mathrm{tr}\, C$. Thus $\mathrm{tr}\, P = |G|^{-1} \sum_g \chi_W(g)\chi_V(g^{-1})$.

N2.2 Because P is idempotent, every A is the sum $P(A) + (A - P(A))$ from image and kernel, with zero intersection. Relative to that decomposition, P has identity and zero blocks, so its trace is the dimension of its image. E1 and RT-02 now give the desired zero or one, and the inverse-trace identity gives the stated conjugated formula.

Export E2/v1. The normalized character inner product equals $\dim \mathrm{Hom}_G(V, W)$ and hence the claimed Kronecker value.

Acceptance gate. Character conjugation belongs on V, the representation appearing with an inverse in the Hom action.

Targeted repair. If the inner product is written without conjugation, restore the inverse representation in the Hom action and recompute its trace. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Unitarize and average on Hom

1. **N1.1.** Start with a positive Hermitian form on each representation and replace it by the sum $\sum_g \langle \rho(g)v, \rho(g)w \rangle$. This remains positive definite and is invariant by reindexing. In orthonormal bases every $\rho(g)$ is unitary, so the trace of its inverse is the complex conjugate of its trace.

2. **N1.2.** The conjugation formula defines a representation on Hom. Reindexing the average shows $P(A)$ is fixed by every group element. A fixed A is unchanged by averaging; fixedness is exactly $A\rho_V(g) = \rho_W(g)A$. Therefore $P^2 = P$ and its image is the intertwiner space.

B. N2 — Compute the trace two ways

1. **N2.1.** For matrices B on W and C on V, the diagonal coefficient of $A \mapsto BAC$ on the matrix unit E_{ij} is $B_{ii}C_{jj}$. Summing over i, j yields $\operatorname{tr}(A \mapsto BAC) = \operatorname{tr} B \operatorname{tr} C$. Thus $\operatorname{tr} P = |G|^{-1} \sum_g \chi_W(g)\chi_V(g^{-1})$.

2. **N2.2.** Because P is idempotent, every A is the sum $P(A) + (A - P(A))$ from image and kernel, with zero intersection. Relative to that decomposition, P has identity and zero blocks, so its trace is the dimension of its image. E1 and RT-02 now give the desired zero or one, and the inverse-trace identity gives the stated conjugated formula.

□

V. WORKED ORCHESTRATION

For $C_3 = \langle g \rangle$, let $\chi_r(g^j) = \omega^{rj}$ for $r = 0, 1, 2$. The inner product of χ_r and χ_s is $(1 + \omega^{s-r} + \omega^{2(s-r)})/3$. It is one when $r = s$ and zero otherwise by the geometric sum identity.

A. Boundary and hypothesis audit

Without complex conjugation, the supposed self-inner-product of χ_1 for C_3 would be $(1 + \omega^2 + \omega)/3 = 0$, although the representation is nonzero irreducible. The conjugation is essential.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 calculation: $\operatorname{tr}(g \cdot A) = \chi_V(g)\chi_W(g)$. **Validation witness:** the zero self-inner-product above. **Diagnosis:** Execution lost the inverse matrix acting on the domain. **Repair:** retain $\rho_V(g)^{-1}$ in the matrix-unit calculation, then use unitarity to conjugate its character. The corrected trace proves the original target; recheck the final orthogonality export.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Derive multiplicities by taking inner products with irreducible characters after complete reducibility is established.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Etingof and collaborators, Introduction to Representation Theory, sections on representations, characters, and $\mathfrak{sl}_2$; MIT Lie Groups and Lie Algebras I lecture notes for solvable Lie algebras.